



National University of Modern Languages

Unity Task 3

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Course:	Computer Graphics
Class:	BSCS VII-A
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1. Objective

The aim of this task was to animate 3D objects within a Unity scene using a custom C# script named ObjectAnimator. This script provides real-time rotation for any attached object and can be customized for individual rotation parameters.

2. Tools and Environment

Software Used: Unity Editor version 6.1.0 (6000.1.5f1)

Target Platform: Windows, Mac, Linux

Development Mode: 3D Project Template

3. Implantation Overview

3.1 Project Setup

Loaded the "Basic3DScene" project in Unity Hub.

Created a new script titled ObjectAnimator.cs and saved it in the Scripts folder.

3.2 Script Functionality

```
using UnityEngine;

public class ObjectAnimator : MonoBehaviour

{

    public Vector3 rotationSpeed = new Vector3(0, 30, 0);

    void Update()

    {

        transform.Rotate(rotationSpeed * Time.deltaTime);

    }

}
```

The script uses the `Update()` method to continuously rotate the object.

3.3 Materials and Textures

- Attached the `ObjectAnimator` script to various 3D objects including cubes, cylinders, and capsules.
- Adjusted the `rotationSpeed` variable in the Inspector to create unique rotation patterns for each object.

3.4 Enhancements

- Applied various textures and materials to the objects to improve visual aesthetics.
- Introduced Unity's `ParticleSystem` for visual effects like smoke and sparkles.
- Integrated audio clips via `AudioSource` components to trigger sound during rotation.
- Tested basic physics interaction by adding `Rigidbody` components to certain objects.
- Explored AR compatibility using Unity AR Foundation for mobile deployment.

3.5 Testing and Optimization

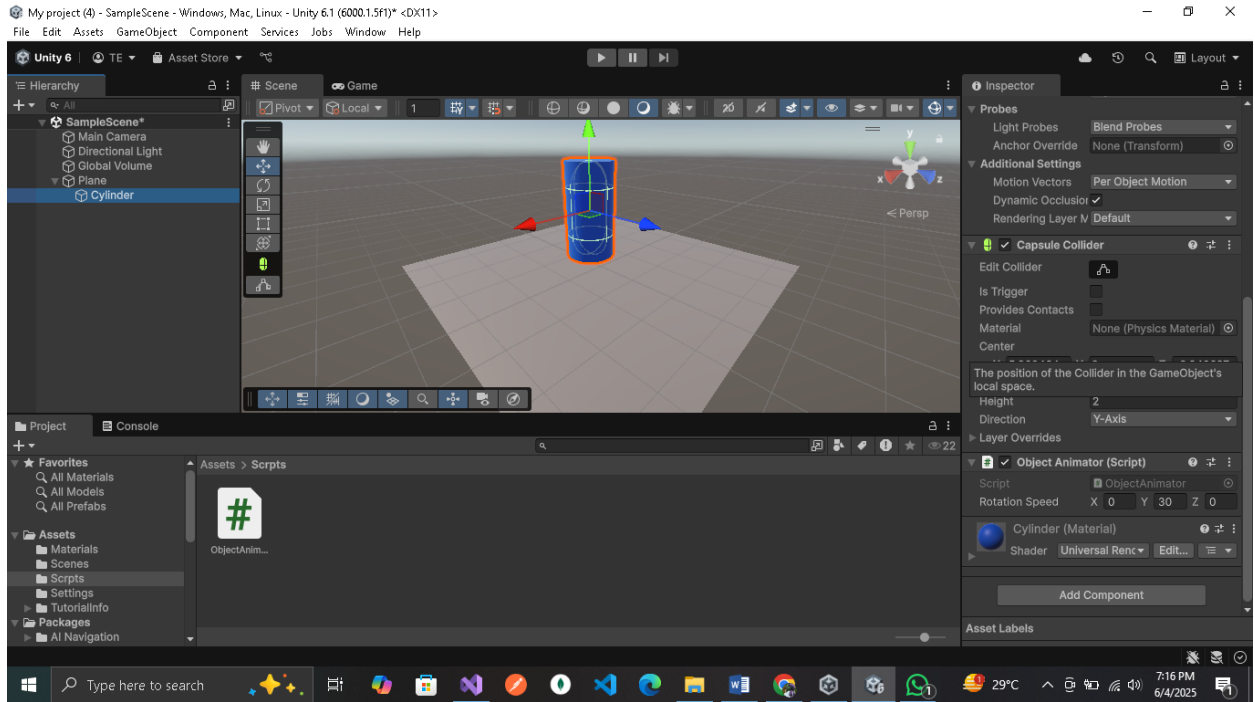
- Used Play Mode to test animation behavior and interactivity.
- Debugged issues by checking variable values in the Inspector.
- Optimized object hierarchy and script performance.

3.6 Reflection:

The `ObjectAnimator` script provided a foundational method for rotating objects in Unity, enhancing the realism and interactivity of the 3D scene. This lab also served as a stepping stone toward more advanced Unity features such as animation, physics integration, and AR development.

4. Screenshot

Below is a screenshot showing the objects



5. Conclusion

This task helped in understanding real-time animation in Unity using scripting. The ObjectAnimator script is flexible and can be used in games, visualizations, and AR projects. Future improvements could involve triggering events based on rotation angles or linking the animation with game logic.
